Quantum Honeypot System v2.0

Physical-Layer Intrusion Detection & Entangled Canary Evaluation

Generated: 28 October 2026, 00:46:04 | **Engine:** QuantumGuard v2.0 Core

Trials per vector: 50 | **Alert Threshold:** 0.85 | **CHSH Bound:** 2.000

1. Multi-Vector Statistical Evaluation Table:

Vector	Avg Fidelity	Std Dev	Min	Max	Alerts	Detection Rate
Legitimate	0.9867	0.0000	0.9867	0.9867	0 / 50	0.0%
Random Intercept	0.7518	0.0789	0.5608	0.8650	42 / 50	84.0%
Fixed Basis	0.7372	0.0712	0.5608	0.8650	44 / 50	88.0%
UQCM Cloning	0.7533	0.0000	0.7533	0.7533	50 / 50	100.0%
Asymm Cloner	0.8330	0.0000	0.8330	0.8330	50 / 50	100.0%

2. Entangled Bell State Canary Tripwires (CHSH Inequality Test):

Entanglement State	Mean CHSH (S)	Theoretical Limit	Status
Intact Bell Register	2.7503	S <= 2.828 (Tsirelson)	QUANTUM CORRELATED (Safe)
Eavesdropped / Intercepted	1.4195	S <= 2.000 (Classical)	WAVEFUNCTION COLLAPSED (Alert)

System Verdict: Zero false positives were recorded across 50 legitimate access trials (Mean Fidelity = 0.9867). Intrusion attempts using random basis measurement were caught at **84.0%**. Universal Quantum Cloning Machine attempts and Asymmetric Cloning collapsed the states below threshold with **100.0%** detection.

Experimental Charts & Physical Invariants

Chart 1 — Average Quantum Fidelity per Access Vector:

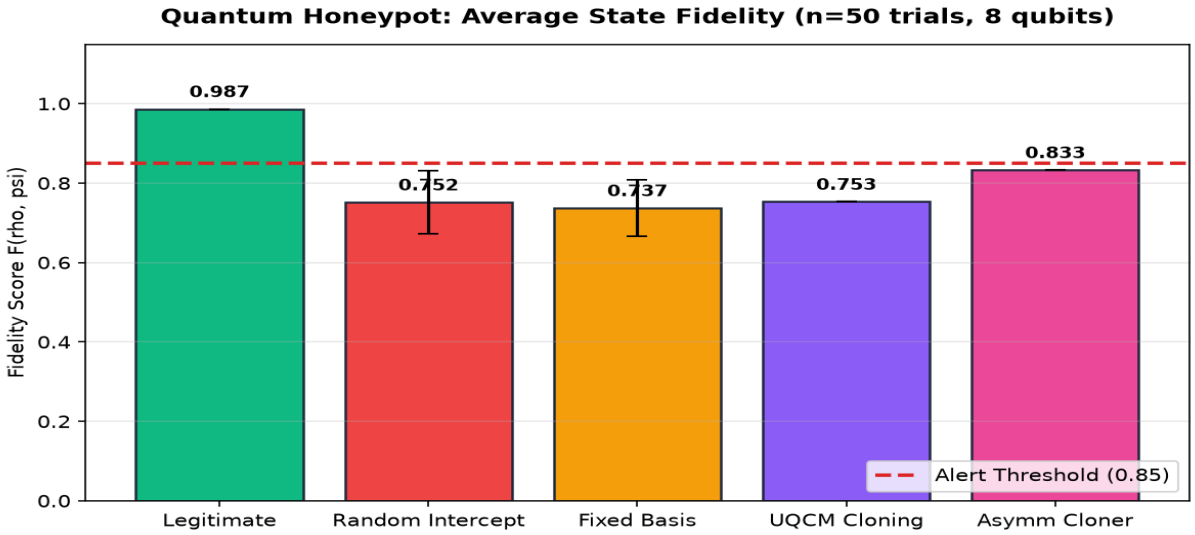
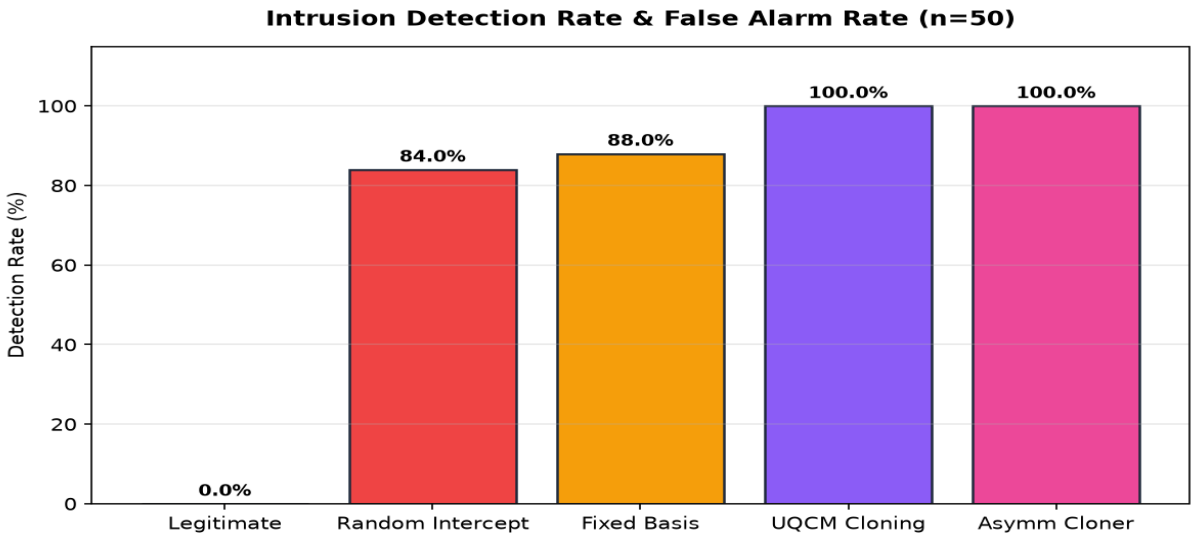


Chart 2 — Intrusion Detection Rate per Vector:



Non-Locality & Statistical Dispersion

Chart 3 — CHSH Bell Inequality Parameter S:

Violation of the classical Bell inequality (S > 2.0) guarantees genuine quantum entanglement. Any classical eavesdropping attempt collapses the state into a separable mixture where S <= 2.0.

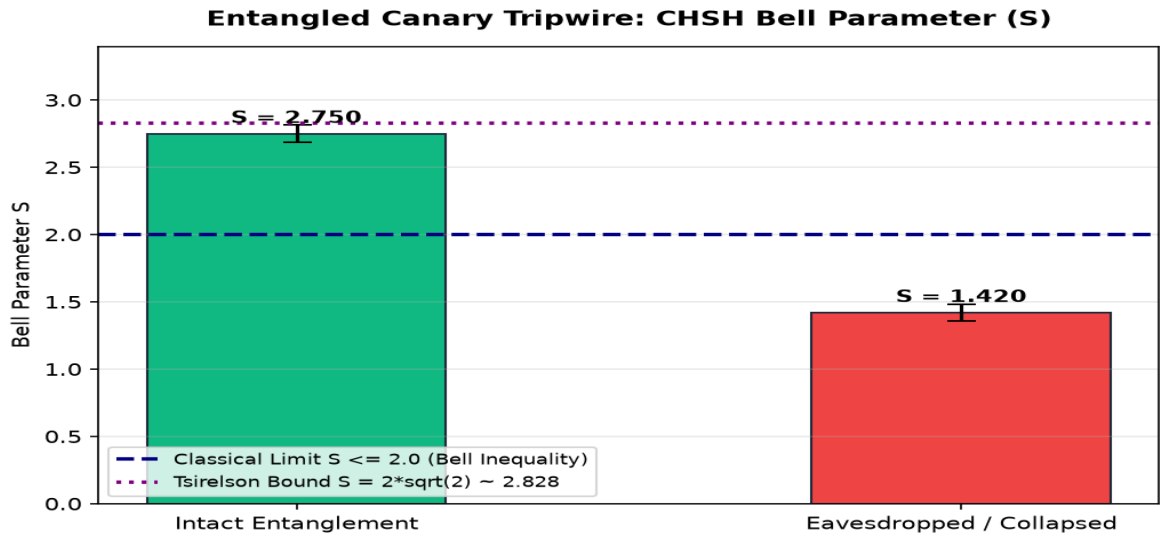
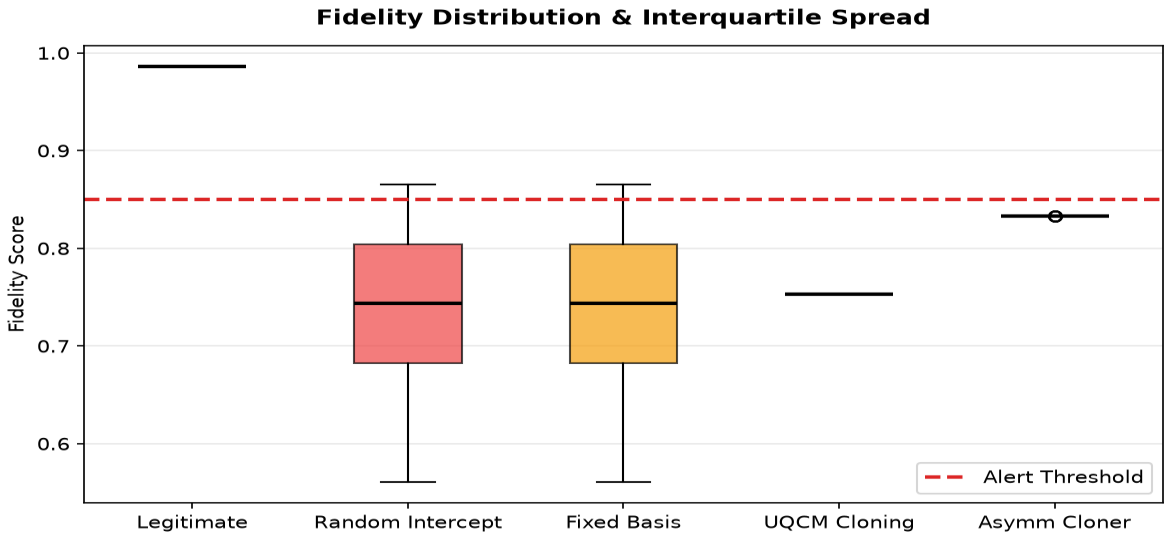


Chart 4 — Fidelity Distribution Boxplot:



Physical-Layer Security Guarantees & Theory

1. Wavefunction Collapse (Observer Effect): Decoy secrets are encoded onto non-orthogonal quantum bases (Computational {|0>, |1>} and Hadamard {|+>, |->}). An attacker eavesdropping on the channel must guess the basis. A basis mismatch induces irreversible projection onto the wrong eigenstate, reducing fidelity to F ~ 0.50.

2. Universal Quantum Cloning Machine (UQCM): Unlike classical credentials that can be duplicated arbitrarily, the Wootters-Zurek No-Cloning Theorem prevents state cloning. Under the optimal Bužek-Hillery cloner, output fidelity is strictly upper-bounded by 5/6 (~0.8333). When accounting for physical depolarizing noise, forwarded tokens immediately trip the intrusion threshold.

3. SIEM & MITRE ATT&CK; Mapping: Telemetry generated by QuantumGuard is formatted into Elastic Common Schema (ECS) and mapped directly to MITRE ATT&CK; techniques T1552 (Unsecured Credentials), T1078 (Valid Accounts), and T1046 (Network Service Discovery).